MODUL PRAKTIKUM 8

Keamanan dan Privasi Big Data

(Studi Kasus: Real-Time Fraud Detection Bank - Spark Structured Streaming + Kafka + Streamlit Dashboard)

Programming Language	: **Python**
Development Environment	: ***Linux Server Environment***
Code Editor	: **VS Code** (*Remote* ***WSL***)
Command Line Interface	: **Bash**
Streaming Ingestion	: **Apache Kafka**
Distributed Data Processing Engine	: **Apache Spark (PySpark)**
Real-Time Visualization	: **Streamlit**
Durasi	: **150 – 180 menit**

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1. TUJUAN PRAKTIKUM

Mahasiswa mampu:

1. Membangun pipeline **Big Data real-time**
2. Menggunakan:
   - Apache Kafka (streaming ingestion)
   - Apache Spark (processing)
   - Streamlit (dashboard)
3. Menerapkan:
   - Data masking
   - Enkripsi data
   - Logging
4. Melakukan **fraud detection real-time**
5. Mengintegrasikan sistem end-to-end

2. KONTEKS INDUSTRI

Dalam industri perbankan modern:

- Transaksi terjadi **ribuan per detik**
- Fraud harus dideteksi **real-time (low latency)**
- Sistem wajib:
  - Secure
  - Scalable
  - Fault-tolerant

📌 Teknologi nyata:
- Kafka → streaming
- Spark → processing
- Data Lake → storage
- Dashboard → monitoring

3. ARSITEKTUR PIPELINE

```
Kafka → Spark Streaming → Secure Processing → Storage → Dashboard
```

3.1. Konsep yang Digunakan

Konsep	Penjelasan
Streaming Processing	Data diproses real-time
Micro-batch	Spark memproses per interval
Data Masking	Menyembunyikan data sensitif
Encryption	Mengamankan data
Fraud Detection	Deteksi transaksi abnormal
Dashboard	Visualisasi real-time
Logging	Audit aktivitas

4. LANGKAH-LANGKAH PRAKTIKUM

📁 A. STRUKTUR PROJECT (WAJIB)

Tambahkan file berikut:

```
scripts/
 ├── kafka_producer_bank.py
 ├── spark_streaming_fraud_v2.py

dashboard/
 ├── fraud_dashboard_v2.py

stream_data/
 ├── realtime_output/

logs/
 ├── fraud_realtime.log
```

⚙️ B. SETUP ENVIRONMENT

✅ **1. Install Java** (Dianjurkan Java 11 lebih stabil) – Jika belum terinstall

```
sudo apt update
sudo apt install openjdk-11-jdk -y
Cek:
java -version
```

✅ **2. Set JAVA_HOME**

```
nano ~/.bashrc
```

Tambahkan:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH
```

Aktifkan:

```
source ~/.bashrc
```

✅ **3. Install Apache Spark** – Jika belum terinstall

```
cd ~
wget https://archive.apache.org/dist/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz
tar -xvzf spark-3.5.1-bin-hadoop3.tgz
sudo mv spark-3.5.1-bin-hadoop3 /opt/spark
```

✅ **4. Set Spark Environment**

Tambahkan ke .bashrc:

```
export SPARK_HOME=/opt/spark
export PYTHONPATH=$SPARK_HOME/python:$SPARK_HOME/python/lib/py4j-*.zip:$PYTHONPATH
export PATH=$SPARK_HOME/bin:$PATH
```

✅ **5. Install Kafka**

```
wget https://archive.apache.org/dist/kafka/3.5.1/kafka_2.13-3.5.1.tgz
tar -xvzf kafka_2.13-3.5.1.tgz
```

⚙️ **C. MENJALANKAN KAFKA**

▶️ **Terminal 1 — Zookeeper**

```
cd kafka_2.13-3.5.1
bin/zookeeper-server-start.sh config/zookeeper.properties
```

▶️ **Terminal 2 — Kafka**

```
cd kafka_2.13-3.5.1
bin/kafka-server-start.sh config/server.properties
```

▶️ **Terminal 3 — Buat Topic**

```
bin/kafka-topics.sh --create \
--topic bank_topic \
--bootstrap-server localhost:9092 \
--partitions 1 \
--replication-factor 1
```

⚙️ **D. KAFKA PRODUCER**

📄 scripts/kafka_producer_bank.py

```
from kafka import KafkaProducer
import json, time, random

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
```

```
    value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

while True:
    data = {
        "nama": random.choice(["Andi","Budi","Citra"]),
        "rekening": str(random.randint(100000,999999)),
        "jumlah": random.randint(100000,100000000),
        "lokasi": random.choice(["Jakarta","Luar Negeri"])
    }

    producer.send("bank_topic", value=data)
    print(data)
    time.sleep(2)
```

E. SPARK STREAMING

scripts/spark_streaming_fraud_v2.py

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import *
from pyspark.sql.types import *

spark = SparkSession.builder.appName("FraudDetection").getOrCreate()

df_kafka = spark.readStream.format("kafka") \
    .option("kafka.bootstrap.servers","localhost:9092") \
    .option("subscribe","bank_topic") \
    .load()

schema = StructType([
    StructField("nama", StringType()),
    StructField("rekening", StringType()),
    StructField("jumlah", IntegerType()),
    StructField("lokasi", StringType())
])

df = df_kafka.selectExpr("CAST(value AS STRING)") \
    .select(from_json(col("value"), schema).alias("data")) \
    .select("data.*")

# Masking
df = df.withColumn("rekening_masked",
    concat(lit("****"), col("rekening").substr(-2,2)))

# Fraud detection
df = df.withColumn("status",
    when(col("jumlah") > 50000000, "FRAUD")
    .when(col("lokasi") == "Luar Negeri", "FRAUD")
    .otherwise("NORMAL"))

# Encryption
```

```
df = df.withColumn("jumlah_encrypted",
    base64(col("jumlah").cast("string")))

query = df.writeStream \
    .format("parquet") \
    .option("path","stream_data/realtime_output/") \
    .option("checkpointLocation","data/checkpoints/") \
    .start()

query.awaitTermination()
```

⚙ F. DASHBOARD STREAMLIT

📄 dashboard/fraud_dashboard_v2.py

```
import streamlit as st
import pandas as pd

st.title("🔒 Real-Time Fraud Detection Dashboard")

df = pd.read_parquet("stream_data/realtime_output/")

st.metric("Total Transaksi", len(df))
st.metric("Total Fraud", len(df[df["status"]=="FRAUD"]))

st.dataframe(df.tail(10))
st.bar_chart(df["status"].value_counts())
```

▶ G. MENJALANKAN SISTEM

Terminal 1

```
python scripts/kafka_producer_bank.py
```

Terminal 2

```
spark-submit \
--packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.1 \
scripts/spark_streaming_fraud_v2.py
```

Terminal 3

```
streamlit run dashboard/fraud_dashboard_v2.py
```

5. CHECKLIST VALIDASI

No	Item	Status
1	Kafka running	☐
2	Producer kirim data	☐
3	Spark streaming aktif	☐
4	Fraud detection berjalan	☐
5	Data tersimpan	☐
6	Dashboard tampil	☐

6. OUTPUT WAJIB

1. Script Python
2. Folder data output
3. Screenshot:
   - Kafka
   - Spark
   - Dashboard
4. Laporan PDF

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7. RUBRIK PENILAIAN

Aspek	1	2	3	4
Streaming	Gagal	Parsial	Jalan	Stabil
Security	Tidak ada	Minim	Cukup	Lengkap
Dashboard	Tidak ada	Basic	Baik	Interaktif
Integrasi	Tidak nyambung	Parsial	Baik	End-to-end
Insight	Dangkal	Cukup	Baik	Mendalam

8. INSIGHT

🔐 **Keamanan**

- Data sensitif harus:
  - Masking
  - Enkripsi

⚡ **Real-Time System**

- Delay = kerugian finansial

🤖 **Fraud Detection**

- Rule-based → awal
- AI → masa depan

💡 **INSIGHT UTAMA**

```
"Dalam sistem Big Data, kecepatan dan keamanan adalah dua hal yang
tidak bisa dipisahkan."
```

🏁 **PENUTUP**

Praktikum ini merepresentasikan:

✅ Sistem real-time industri
✅ Pipeline Big Data lengkap
✅ Fraud detection system